Sign Misattribution in Step-Independent Agent RL: Dual Mechanism Diagnosis and Episode-Level Correction

Anonymous ACL submission

Abstract

Reinforcement learning for multi-turn LLM agents requires credit assignment across long interaction trajectories. The dominant training paradigm decomposes these trajectories into fixed-size windows optimized independently via Group Relative Policy Optimization (GRPO), yet this ubiquitous design choice has received little scrutiny. We identify a structural credit assignment pathology driven by two distinct mechanisms: *policy information loss*, which truncates cross-window context and imposes a constant performance ceiling, and *advantage sign misattribution*, which causes the majority of gradient updates for locally correct actions to carry incorrect sign. A factorial experiment isolates each mechanism: removing sign misattribution converts catastrophic collapse into bounded learning; removing information loss enables out-of-distribution (OOD) generalization. Direct sign measurement confirms the theoretical prediction, with credit agreement dropping to near-random levels. Episode-level GRPO, which treats the full trajectory as a single optimization sample, eliminates both failure modes. However, a systematic multi-seed evaluation reveals that avoiding collapse is necessary but not sufficient: 73% of seeds stabilize, yet only 36% achieve statistically significant OOD transfer. Cross-scale verification at 8B parameters and cross-environment validation on ScienceWorld provide preliminary evidence that the pathology extends beyond our primary setting.

1 Introduction

Reinforcement learning from sparse terminal reward has emerged as a promising paradigm for training LLM agents on multi-turn interactive tasks spanning 30–50 interaction steps (Kim et al., 2025; Bai et al., 2024; Wang et al., 2022). Group Relative Policy Optimization (GRPO) (Shao et al., 2024), with its critic-free design, has become the algorithm of choice. Yet a gap persists between the promise of RL-driven agent improvement and the reality of training outcomes: recent agent RL systems (Wang et al., 2025, 2026a; Qin et al., 2026; Feng et al., 2025) share a common, unscrutinized implementation pattern (decomposing trajectories into fixed-size windows, each optimized independently) and frequently report training instability without diagnosing its root cause.

We show this *step-independent* default introduces a fundamental credit assignment pathology. Step-independent training partitions trajectories into non-overlapping h-step windows, each optimized via GRPO. This produces a two-phase collapse: task success drops to 0% while action validity initially remains high, then continued training destroys even basic formatting. Standard remedies (dense reward shaping, discount tuning) fail. Concurrent methods that improve per-step credit quality (GiGPO (Feng et al., 2025), GAGPO (Zhu et al., 2026), Turn-PPO (Li et al., 2026a), HCAPO (Tan et al., 2026)) address the symptom through improved credit assignment mechanisms, validating that sign correctness is the critical factor. Our contribution is orthogonal: we formally diagnose *why* the default windowing fails.

We trace the collapse to two mechanisms at different layers. ***Policy information loss*** (M1) operates at the inference layer: each window lacks access to earlier observations, imposing a constant performance ceiling. ***Advantage sign misattribution*** (M2) operates at the training layer: under binary reward, the advantage sign reflects episode success rather than local action quality. Our theoretical analysis shows that the majority of gradient updates for locally correct actions carry incorrect sign (Section 3), a prediction confirmed by direct measurement (Section 4.4). Sign errors, amplified by the M1–M2 feedback loop, cause progressive degradation rather than merely slow convergence. Our primary contribution is diagnostic: we identify, formalize, and empirically validate a previously

uncharacterized pathology in the dominant training paradigm. Episode-level GRPO (not a new method, but the original GRPO design) eliminates both mechanisms by treating the full trajectory as a single optimization sample, achieving substantial OOD improvement where step-independent collapses to 0%, with effects persisting at 8B scale.

Our contributions:

1. **Dual mechanism diagnosis.** We formalize *information loss* and *sign misattribution* as distinct failure modes with different temporal behaviors (constant ceiling versus cumulative degradation) and derive closed-form expressions for the probability of incorrect gradient direction.
2. **Empirical separation.** A factorial experiment isolates each mechanism, direct sign measurement confirms the theoretical prediction at near-random levels, and preliminary cross-scale (8B) and cross-environment (ScienceWorld) evidence supports generality of the pathology.
3. **Episode-level fix: necessary but not sufficient.** GRPO's original episode-level design addresses both mechanisms, but a systematic multi-seed evaluation reveals that most seeds avoid collapse while only a minority achieve statistically significant OOD generalization. We analyze this seed-dependent transfer gap.

2 Method

2.1 Problem Formulation and GRPO

We model the multi-turn agent as a partially observable Markov decision process (POMDP) where the policy $\pi_\phi(a_t \mid h_t)$ generates actions conditioned on interaction history $h_t = (o_1, a_1, \ldots, o_t)$. Episodes of T steps terminate with binary reward $R \in \{0, r_+\}$. GRPO (Shao et al., 2024) samples K rollouts per task and computes group-relative advantages:

$$\hat{A}_k = \frac{R_k - \mu}{\sigma + \epsilon}, \qquad \mu = \frac{1}{K}\sum_{j=1}^{K} R_j, \quad \sigma = \sqrt{\frac{1}{K}\sum_{j=1}^{K}(R_j - \mu)^2} \tag{1}$$

GRPO is episode-level by design: each rollout receives one return and one advantage, with no inherent temporal discounting.

2.2 Step-Independent Windowing

Agent RL frameworks must reconcile GRPO's episode-level design with the memory constraints of long trajectories. RAGEN (Wang et al., 2025) introduces **step-independent training**: the trajectory is partitioned into non-overlapping h-step windows, each receiving a discounted return:

$$R_k = \gamma^{(N-k)} \cdot R_{\text{episode}} \tag{2}$$

where $N = \lceil T/h \rceil$ is the number of windows. Each window is optimized as an independent GRPO sample. This reduces sequence length from $O(T)$ to $O(h)$ and multiplies the effective batch size, but structurally alters credit assignment.

2.3 Dual Mechanism Analysis

Two mechanisms jointly cause training failure under step-independent optimization.

2.3.1 Mechanism 1: Policy Information Loss

Step-independent training truncates context at both inference and training time. The policy $\pi_\phi(a_t \mid h_{\text{window}_k})$ conditions on only the local h-step history, losing access to earlier observations. With h=2, a window at step 8 has no access to the outcome of an exploration action at step 4, so the policy cannot learn strategies spanning more than h steps. This imposes a *constant* performance ceiling: no amount of optimization can recover absent information.

2.3.2 Mechanism 2: Advantage Sign Misattribution

Group normalization under step-independent training produces advantages whose sign reflects the *episode* outcome, not local action quality:

$$\text{sign}(\hat{A}_k) \approx \text{sign}(R_{\text{episode}} - \mu) \tag{3}$$

In failed episodes ($R_{\text{episode}} = 0$), all windows receive negative advantages when $\mu > 0$, including windows with locally correct actions. Conversely, successful episodes reinforce all windows, including those with suboptimal actions. Unlike M1's constant ceiling, sign misattribution is *cumulative*: each wrong-sign gradient step further erodes policy competence. The discount factor $\gamma^{(N-k)}$ introduces magnitude decay across positions, but sign errors are strictly more harmful: magnitude decay slows convergence, while sign errors cause active divergence.

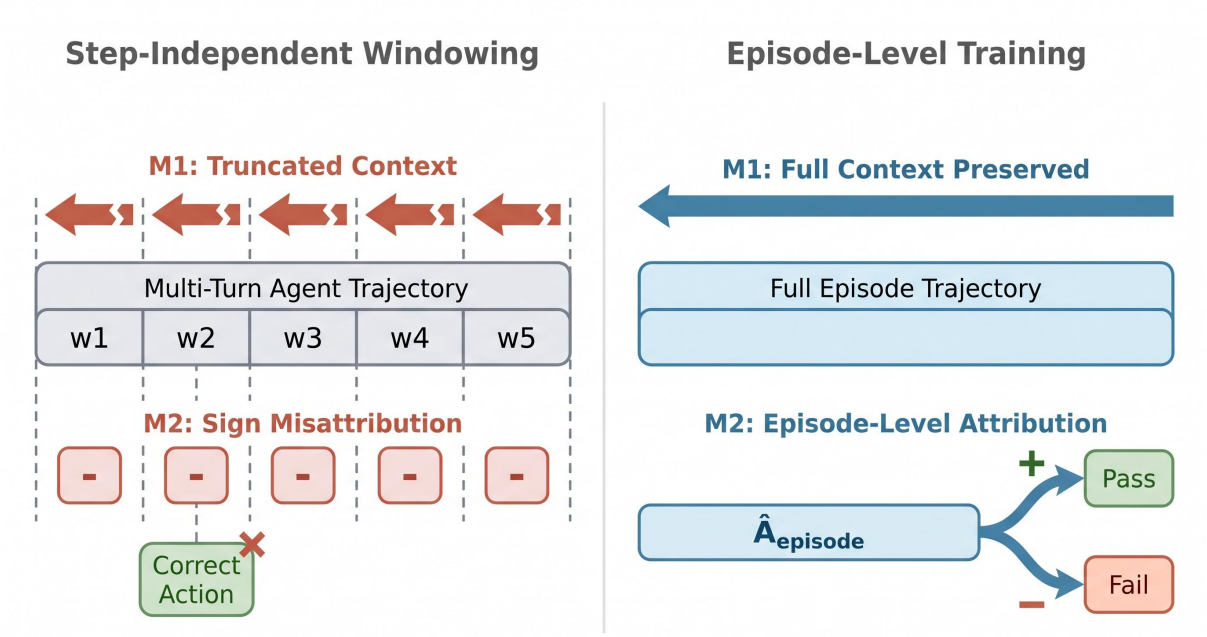


Figure 1: Step-independent windowing (left) partitions the trajectory into fixed windows, each optimized independently. This introduces two failure mechanisms: M1 (information loss) truncates context at window boundaries, and M2 (sign misattribution) assigns all windows the same advantage sign based on episode outcome. Episode-level training (right) treats the full trajectory as a single sample, eliminating both mechanisms.

2.4 Episode-Level Training

Episode-level training addresses both mechanisms by treating the complete trajectory as a single GRPO sample. The full history h_t is available at every step (eliminating M1), and a single advantage $\hat{A}$ applies uniformly to all tokens (eliminating M2). Uniform reinforcement of successful trajectories preserves good sub-strategies; window-level sign errors actively destroy them. This is GRPO's original design; step-independent windowing is the deviation.

2.5 Factorial Experiment Design

To separate the two mechanisms, we implement a 2×2 factorial varying observation scope and advantage granularity: **Cell A** (truncated h=2 + step-level advantage; both mechanisms active), **Cell B** (truncated h=2 + episode-level advantage; only M1 active), and **Cell D** (full trajectory + episode-level advantage; neither active). The fourth cell, **Cell C** (full trajectory + step-level advantage; only M2 active), would isolate sign misattribution without information loss. We attempted this configuration but encountered persistent training instability (Section 4.2; Appendix D). Comparing A vs. B isolates sign misattribution; comparing B vs. D isolates information loss.

3 Theoretical Analysis

Sign misattribution is the more fundamental failure mechanism because it causes active divergence, not merely slow convergence. We derive the probability of incorrect gradient direction below; full proofs appear in Appendix A.

Definition 1 (Locally correct action). *An action a_t is **locally correct** if it would receive positive advantage under an oracle that observes the full episode history h_t and has access to the true action-value function. Equivalently, window k contains locally correct actions if replacing them (holding all other windows fixed) would decrease the probability of episode success.*

Proposition 1. *Under step-independent training with binary episode reward $R_{episode} \in \{0, r_+\}$, the probability that window k's advantage $\hat{A}_k$ has the correct sign (aligned with local action quality) approaches the base success rate p as the number of rollouts K per group increases.*

Proof sketch. In a failed episode ($R_{\text{episode}} = 0$), all windows receive $R_k = 0$. If any rollout in the group succeeded, $\mu > 0$, so every window in the failed episode receives $\hat{A}_k < 0$—including windows with locally correct actions. The probability of receiving the wrong sign in a failed episode is $1 - (1 - p)^{K-1}$ (at least one group member suc-

ceeded). As K grows, this approaches $1-p$, and the overall probability of correct sign approaches p. □

The key distinction is temporal. Information loss (M1) imposes a constant ceiling determined by h; sign misattribution (M2) is cumulative—each wrong-sign gradient step erodes policy competence below that ceiling. A wrong-sign gradient moves the optimizer *away* from good policies; a reduced-magnitude gradient from discounting ($\gamma^{(N-k)}$) still points toward good policies. Sign errors produce active divergence; magnitude errors merely slow convergence (Appendix A.1).

Proposition 2. *Let p denote the episode success rate and K the number of rollouts per group. Under step-independent training with binary episode reward, the fraction of gradient updates with correct sign is:*

$$P(\text{correct} \mid \text{update}) = \frac{p}{1-(1-p)^K} \qquad (4)$$

Derivation. Three mutually exclusive cases determine gradient direction for a window with locally correct actions: (1) episode succeeded (probability p): correct sign; (2) episode failed but group contains a success (probability $(1-p)[1-(1-p)^{K-1}]$): wrong sign; (3) entire group failed (probability $(1-p)^K$): zero gradient. Conditioning on non-zero gradient (Cases 1 and 2) yields Equation 4.

At p=12.5%, K=8, only 19.1% of gradient updates have the correct sign (Figure 2). In the large-K limit, $P(\text{correct} \mid \text{update}) \to p$: ∼87.5% of updates for locally correct windows point in the wrong direction.

Severity scales with task difficulty. Sign misattribution becomes *more* severe as the task gets harder (Equation 4): at p=50%, $P(\text{correct} \mid \text{update}) = 50.2\%$—barely above random; at p=5%, only 5.1% of updates are correct. Multi-turn agent tasks with low baseline success rates are precisely where credit assignment matters most, yet where step-independent training produces the worst gradient signal. Increasing K does not help: more updates occur (fewer all-fail groups), but each remains equally likely to carry the wrong sign.

Reward saturation. When all K rollouts receive the same binary reward, $\sigma = 0$ and $\hat{A}_k = 0$, producing zero gradient. At p=12.5%, K=8, the all-fail probability is $(1-p)^K \approx 34.4\%$; combined with the negligible all-success case ($p^K < 10^{-7}$),

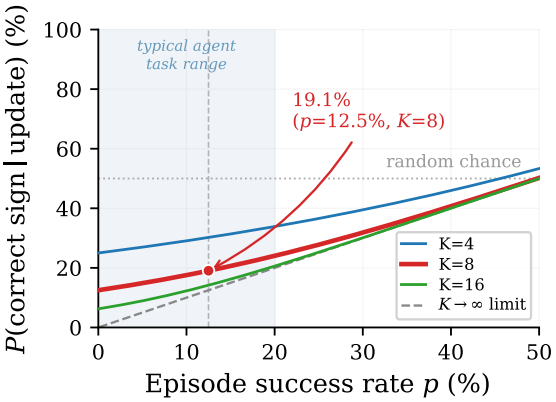


Figure 2: Probability of correct gradient sign as a function of episode success rate p (Proposition 2). At typical agent task success rates ($p < 0.2$, shaded), the majority of gradient updates carry incorrect sign regardless of group size K. At ALFWorld's baseline (p=12.5%, K=8), only 19.1% of updates are correct. The dashed diagonal shows the large-K limit $P_{\text{correct}} \to p$.

roughly one-third of batches yield no gradient at all. This compounds sign misattribution: of the ∼66% of batches that produce a gradient, only 19.1% have the correct sign. The effective learning rate is $0.66 \times 0.19 \approx 12.5\%$ of nominal—well below the regime where convergence guarantees hold.

Cumulative divergence. Sign misattribution produces a systematic gradient bias. Let θ^* denote an optimal policy parameterization. At each update, the expected gradient under step-independent training is:

$$\mathbb{E}[\nabla_\theta \mathcal{L}] = P_c \cdot g_+ + (1-P_c) \cdot g_- \qquad (5)$$

where $P_c = P(\text{correct} \mid \text{update})$, g_+ points toward θ^*, and g_- points away. Empirical measurement from training logs confirms approximately comparable per-group advantage magnitudes ($|\hat{A}_{\text{correct}}|/|\hat{A}_{\text{wrong}}| \approx 1.32$). When $P_c < 0.5$ (which holds for $p < 0.5$; Equation 4), the expected gradient points weakly away from good policies. The cumulative effect of this directional bias, amplified by the M1–M2 positive feedback loop (Section 2.3), drives progressive policy degradation: information loss (M1) caps p; lower p increases wrong-sign frequency under M2, which further depresses p, producing the empirically observed collapse.

4 Experiments

Step-independent training collapses while episode-level training succeeds; a factorial design separates the two mechanisms; and window size h modulates severity.

4.1 Setup

We evaluate on ALFWorld (Shridhar et al., 2021), a text-based household environment with six task types requiring sequential reasoning over up to 50 steps. Reward is binary: $R \in \{0, r_+\}$. The base policy is Qwen3-1.7B fine-tuned on expert demonstrations (SFT baseline: 12.5% in-distribution). At this baseline success rate, Proposition 2 predicts the majority of gradient updates carry incorrect sign. All conditions use RAGEN (Wang et al., 2025) (v0.1, commit `3a6b48b`)[1] with GRPO (Shao et al., 2024) (K=8; full hyperparameters in Appendix B). We report success rate on a 134-task OOD set (primary) and 32-task in-distribution set (secondary). Seed counts are noted per condition.

4.2 Factorial Mechanism Separation

We implement the factorial design from Section 2.5 with two seeds per condition. Cell C (full observation, step-level advantage) would isolate M2 without M1, but we encountered persistent training instability in this configuration (Appendix D).

4.3 Results

Cell A collapses in two phases (Table 1): task success drops to 0% while action validity initially remains high, then continued sign misattribution destroys even basic formatting.Cell B eliminates sign misattribution while retaining information loss: training success improves, but OOD generalization is bounded by the information-loss ceiling. Both seeds converge to the same OOD ceiling despite different training trajectories, confirming the bottleneck is architectural rather than statistical.

Cell D is the only condition that substantially exceeds the SFT baseline, with monotonic OOD improvement across both seeds.

The A→B comparison isolates M2: eliminating sign misattribution converts catastrophic collapse into bounded learning. The B→D comparison isolates M1: removing the information-loss ceiling enables OOD generalization. Both mechanisms are necessary to explain the full failure pattern. Figure 3(a) visualizes these trajectories.

Table 1: 1.7B results on ALFWorld (best checkpoint per seed, evaluated on held-out test sets). In-Dist: 32-task set; OOD: 134-task set. Cell A reports best seed; Cell B reports best checkpoint (step 5). Cell D reports mean of two seeds (In-Dist range: 28.1–40.6; OOD range: 10.4–23.9).

Condition	Mechanisms	In-Dist (%)	OOD (%)
SFT baseline	—	12.50	6.72
Cell A (h=2, step)	M1+M2	0	0
Cell B (h=2, episode)	M1 only	18.75	5.97
Cell D (full, episode)	neither	**34.38**	**17.16**
Dense reward	M1+M2	0	0
REINFORCE++	neither	12.50	—

Additional conditions. Dense reward, REINFORCE++, and discount-factor sweep experiments all confirm that sign misattribution, not reward sparsity or discounting, drives the failure (Appendix D).

Window-size sweep. Varying window size reveals a credit assignment threshold between h=10 and h=20 that separates collapse from recovery (Figure 3b; seed 1). Below this threshold, all conditions collapse to 0%; above it, success rate increases monotonically with window size. This threshold is observed in a single seed; multi-seed verification is needed to confirm its robustness.

4.4 Sign Misattribution Verification

We operationalize local correctness (Definition 1) through episode outcome: in a successful episode, all windows with valid actions are labeled locally correct. The credit agreement rate (the fraction of such windows where GRPO's advantage sign is positive) is 52% under step-independent training (h=2, n=3,000 windows), marginally above random chance. Advantage sign correlates near-perfectly with episode outcome but shows near-zero dependence on local action quality. Combined with γ=1.0 collapse (Appendix D), these measurements confirm sign misattribution as the causal driver. This operationalization is a conservative approximation: successful episodes may contain suboptimal windows, making our label an upper bound on true local correctness. Even under this generous criterion, credit agreement remains near-random, strengthening the diagnosis.

Under pure binary reward, sign-episode correlation reaches 100%, directly confirming M2: advantage sign perfectly tracks episode outcome (Fig-

[1] Our implementation extends RAGEN v0.1 with episode-level training support. Code: https://github.com/RAGEN-AI/RAGEN.

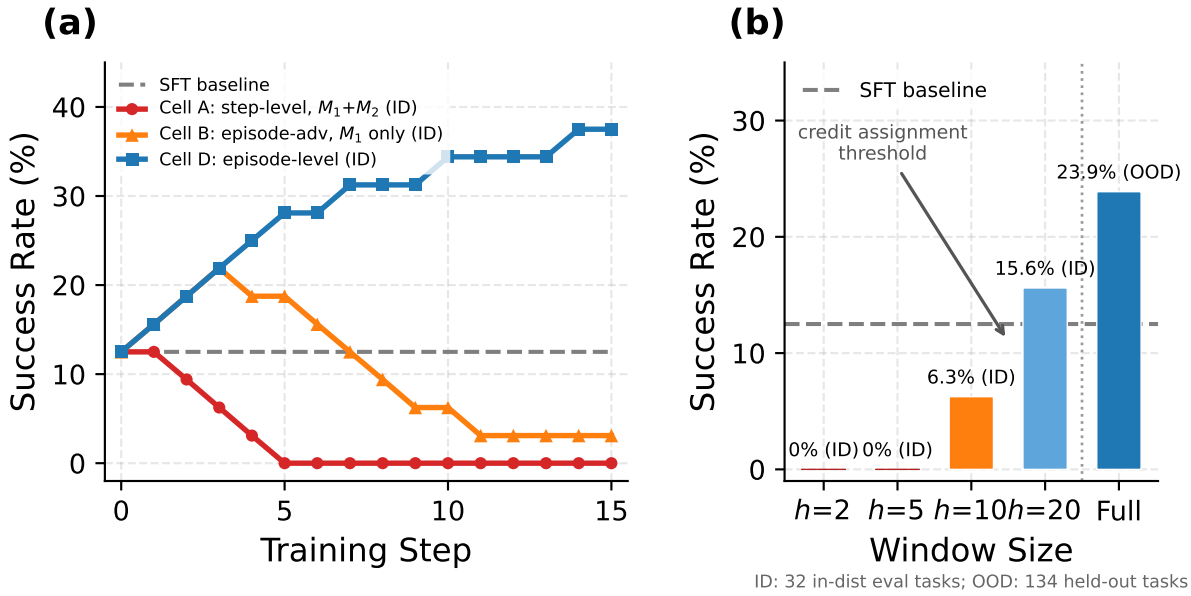


Figure 3: (a) Training dynamics across factorial conditions (metric type indicated in legend). Cell A collapses within 5 steps; Cell B shows bounded learning; Cell D improves monotonically. (b) Window size h vs. success rate (seed 1). $h \leq 20$ bars use a 32-task in-distribution eval set; the episode-level bar uses a 134-task OOD set (dotted separator); magnitudes are not directly comparable across eval sets. A credit assignment threshold between h=10 and h=20 separates collapse from recovery. Hatched bars indicate collapsed conditions.

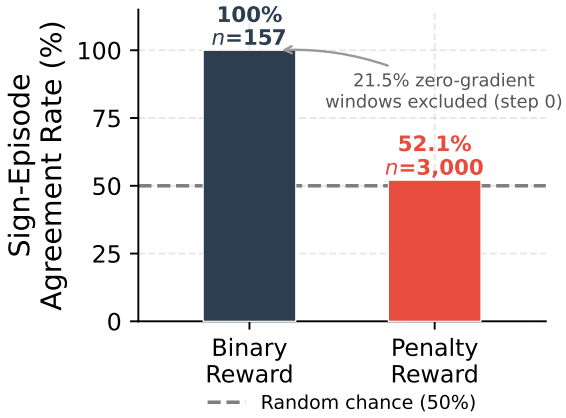


Figure 4: Sign-episode agreement rate under binary reward (penalty=0) vs. penalty reward. Under binary reward, advantage sign perfectly tracks episode outcome (100%, n=157), directly confirming M2. Penalty reward reduces this to 52.1% (n=3,000), near random chance (dashed line).

ure 4). Penalty reward partially breaks this by introducing within-group variance, but credit agreement remains near-random for local quality.

4.5 Episode-Level Training: Seed Variability

Episode-level GRPO addresses the dual-mechanism failure, but training outcomes remain seed-dependent. We systematically evaluate N=11 random seeds under identical hyperparameters, revealing a 2×2 structure: *training stability* (collapsed vs. non-collapsed) × *OOD generalization* (stagnant vs. generalized).

We define **collapsed** seeds as those where in-distribution success declines >50% from peak and remains below baseline for ≥3 consecutive steps. **Non-collapsed** seeds are the complement. Among non-collapsed seeds, we further distinguish **generalized** (Fisher exact p<0.01 vs. SFT on the 134-task OOD set) from **stagnant** (not significant vs. SFT). Table 2 reports results for each seed at its best-performing checkpoint.

Three distinct outcomes emerge (Table 2): roughly a quarter of seeds collapse with response length inflation (Appendix E), a third are stagnant with OOD performance indistinguishable from SFT, and a third generalize with significant OOD improvement. No collapsed seed generalizes, and training duration does not explain the gap between stagnant and generalized seeds. Stable training is necessary but not sufficient; the mechanism driving seed-dependent transfer remains an open question (Section 6).

Table 2: Episode-level GRPO across 11 seeds (134 OOD tasks, τ=0.4, best checkpoint per seed). Category: C=Collapsed, S=Non-collapsed (stagnant), G=Generalized (Fisher exact $p<0.01$ vs. SFT). Wilson 95% CI. SFT baseline: 6.72% (9/134). Seed2 achieves $p=0.03$ (Fisher exact, two-sided), falling outside the conservative $p<0.01$ threshold, and is classified as Stagnant. Step-5 results for all seeds in Appendix F.

Seed	Step	Cat.	OOD (%)	95% CI	vs SFT
seed2048‡	5	C	8.21	[4.6, 14.2]	NS
seed42‡	5	C	12.69	[8.0, 19.4]	NS
seed8	5	C	11.94	[7.5, 18.5]	NS
seed200	5	S	10.45	[6.3, 16.8]	NS
seed300	5	S	10.45	[6.3, 16.8]	NS
seed100	5	S	11.19	[6.9, 17.6]	NS
seed100	10	S	12.69	[8.0, 19.4]	NS
seed2‡	15	S	15.67	[10.5, 22.8]	$p=0.03$
seed1	5	G†	23.88	[17.4, 31.8]	$p<0.001$
seed5	10	G	29.10	[22.1, 37.3]	$p<0.001$
seed6	15	G	29.10	[22.2, 37.1]	$p<0.001$
seed4	15	G	30.60	[23.5, 38.7]	$p<0.001$

† RAGEN built-in eval; checkpoint unavailable for re-evaluation.
‡ OOD evaluated via RAGEN built-in eval; not independently re-evaluated under D134 protocol.
All results are single-run evaluations at τ=0.4; see Limitations for eval variance discussion.

4.6 Cross-Scale and Cross-Environment Verification

A single-seed experiment on Qwen3-8B provides preliminary evidence that both failure and fix persist at scale. Cell A collapses within two steps; Cell D reaches 90.3% OOD, substantially exceeding the SFT baseline (full results in Appendix C). Restricting context to h=2 produces a large M1 gap, confirming that information loss remains severe at scale. Single-seed due to memory constraints.Step-independent training also collapses on ScienceWorld (Wang et al., 2022): Cell A drops to 0% throughout all 15 steps. Restricting context to h=2 produces a comparable M1 gap to the 8B ALFWorld setting, suggesting the pathology is environment-general.

5 Related Work

RL for multi-turn LLM agents. PPO (Schulman et al., 2017), GRPO (Shao et al., 2024), and DPO (Rafailov et al., 2023) underpin RLHF (Ouyang et al., 2022) and have been extended to multi-turn agents (Yao et al., 2023) through step-independent windowing (Wang et al., 2025). Subsequent work improves algorithms (GiGPO (Feng et al., 2025)), reward design (Self-Guide (Wang et al., 2026a)), or pipelines (SPEAR (Qin et al., 2026), DigiRL (Bai et al., 2024)), but all assume trajectory decomposition into fixed windows is benign. We show this assumption is false.

Credit assignment. TD(λ) (Sutton, 1988), GAE (Schulman et al., 2016), and REINFORCE (Williams, 1992) span bootstrapping to Monte Carlo returns; reward shaping (Ng et al., 1999) and counterfactual baselines (Foerster et al., 2018) address sparse reward. Several studies corroborate our diagnosis: IRC (Modecrua et al., 2026) and MC-GRPO (Kim, 2025) observe sign disagreement between per-turn and outcome advantages, GRPO-λ (Parthasarathi et al., 2025) proves estimation error grows with position, and OAR (Li et al., 2026b) identifies reward aliasing. Turn-PPO (Li et al., 2026a), GAGPO (Zhu et al., 2026), MT-GRPO (Wei et al., 2025), SWEET-RL (Zhou et al., 2025), and iStar (Liu et al., 2025) propose finer-grained credit methods; HCAPO (Tan et al., 2026) validates that sign correctness is the critical bottleneck. We diagnose *why* the default fails, informing when finer-grained credit is safe.

GRPO collapse phenomena. RAGEN-2 (Wang et al., 2026b) proves advantage collapse when reward variance vanishes, GRPO-Collapse (Deng et al., 2025) identifies a death spiral in Search-R1, and RC-GRPO (Zhong et al., 2026) and EDGE-GRPO (Zhang et al., 2025) address variance collapse. Our dual-mechanism diagnosis unifies these: reward variance collapse is a special case of reward saturation (Section 3), and the death spiral follows from the M1–M2 vicious cycle.

6 Discussion

Generality and mechanism interaction. Sign misattribution arises whenever three structural properties co-occur: (a) step-level windowing, (b) sparse binary reward, and (c) group-relative normalization. Under these conditions, the advantage sign collapses to a proxy for episode outcome, and severity scales with task difficulty ($P(\text{correct} \mid \text{update}) \to p$). M1 and M2 interact in a vicious cycle: M1 lowers p, increasing wrong-sign updates under M2, which further degrades p. The A→B comparison provides direct evidence: identical observation scope (h=2), yet eliminating step-level advantage converts catastrophic collapse into

bounded learning. Six factors explaining why step-independent became the default are detailed in Appendix A.3.

Seed-dependent OOD generalization. The multi-seed evaluation reveals a bimodal distribution: a stagnant cluster with OOD performance indistinguishable from SFT and a generalized cluster with significant improvement. Training duration does not explain the gap, pointing to initialization sensitivity as the likely driver.

Practitioner guidance. Use episode-level credit when the full episode fits in context; otherwise use the largest feasible h ($h \geq 20$; $h \leq 10$ collapses). Finer-grained methods should be applied only after establishing an episode-level baseline.

7 Conclusion

Step-independent training fails through two mechanisms: information loss imposes a constant performance ceiling while sign misattribution causes cumulative degradation, with the majority of gradient updates carrying incorrect sign. Episode-level GRPO addresses both failure modes, achieving substantial OOD improvement at both 1.7B and 8B scales, though only a minority of seeds achieve statistically significant generalization. Sign errors are more destructive than magnitude imprecision; this finding applies to any agent RL method combining step-level windowing, sparse binary reward, and group-relative normalization.

Limitations

Our experiments span two environments (ALFWorld, ScienceWorld) and two scales (1.7B, 8B); additional environments and scales would strengthen generality. The 8B and ScienceWorld results report single-seed runs due to computational constraints. Among 11 seeds at 1.7B, only 36% achieve statistically significant OOD generalization; we cannot rule out that stagnant seeds would eventually generalize with substantially more training steps, though seed100's minimal improvement from step 5 to step 10 suggests the bottleneck is not simply training duration. The mechanism driving seed-dependent transfer remains an open question. All OOD evaluations use temperature $\tau = 0.4$ with a single run per checkpoint; repeated evaluations of the same checkpoint show non-trivial variance (e.g., 20pp range for seed5), exceeding binomial predictions and reflecting stochastic decoding effects. For very long episodes ($T > 100$) or tasks with many independent sub-goals, uniform episode-level advantage may limit efficiency even though it preserves direction correctness.

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A Theoretical Analysis: Full Derivations

A.1 Why Sign Misattribution is the Core Mechanism

The two mechanisms identified in Section 2.3 operate at different layers and produce qualitatively different failures. Policy information loss (Mechanism 1) operates at the inference layer, restricting the observable context and establishing a structural performance ceiling that is constant across training. Sign misattribution (Mechanism 2) operates at the training layer, corrupting gradient directions and progressively degrading performance below this ceiling.

Proof of Proposition 1. Define "correct sign" as follows: for a window containing locally beneficial actions, the correct sign of $\hat{A}_k$ is positive (reinforcement); for a window containing locally detrimental actions, the correct sign is negative (suppression). Under binary reward, the return for window k is $R_k = \gamma^{(N-k)} \cdot R_{\text{episode}}$ (Equation 2). In a failed episode ($R_{\text{episode}} = 0$), all windows receive $R_k = 0$ regardless of their local action quality. If the group contains at least one successful rollout, the group mean $\mu > 0$, and all windows in the failed episode receive negative advantages after normalization via Equation 1. This includes windows where the agent took locally correct actions. Now consider a window with locally correct actions. In a successful episode (probability p), the window receives positive $\hat{A}_k$, which is the correct sign. In a failed episode (probability $1-p$), the window receives negative $\hat{A}_k$ whenever $\mu > 0$, which is the wrong sign. The probability that $\mu = 0$ (all K rollouts in the group failed) is $(1-p)^K$, which shrinks exponentially with K. As K grows, nearly every failed episode produces a non-zero group mean, and the window receives the wrong sign with probability approaching $1-p$. The probability of correct sign therefore approaches p. □

The distinction between sign error and magnitude error further clarifies why Mechanism 2 is catastrophic while Mechanism 1 is merely suboptimal. The discount factor $\gamma^{(N-k)}$ in Equation 2 introduces magnitude decay across window positions: for an episode with $T = 50$ steps and $h = 2$, the first window ($k = 1$) retains $\gamma^{N-1} = 0.95^{24} \approx 0.29$ of the terminal window's return. This attenuation reduces the gradient magnitude for early windows but preserves the gradient direction. A gradient with correct sign and reduced magnitude produces slow convergence, a condition recoverable through longer training. A gradient with wrong sign produces active divergence: the optimization pushes the policy away from good behaviors and toward bad ones.

A.2 Formal Probability Analysis

Derivation of Proposition 2. For a window with locally correct actions, the gradient direction depends on $\text{sign}(\hat{A}_k)$, which depends on whether the episode succeeded and whether the group contains any successful rollouts. Three mutually exclusive cases arise.

Case 1: Episode succeeded (probability p). The window receives $R_k = \gamma^{(N-k)} \cdot r_+ > 0$. With at least one failed rollout in the group, $R_k > \mu$ after normalization, producing $\hat{A}_k > 0$.[2] The gradient reinforces the window's actions. For a window with locally correct actions, this is the correct direction.

Case 2: Episode failed, group contains at least one success (probability $(1-p)[1-(1-p)^{K-1}]$). The window receives $R_k = 0 < \mu$, producing $\hat{A}_k < 0$. The gradient suppresses the window's actions. For a window with locally correct actions, this is the wrong direction.

Case 3: Episode failed, entire group failed (probability $(1-p)\cdot(1-p)^{K-1} = (1-p)^K$). All returns are zero, so $\mu = 0$ and $\sigma = 0$. The advantage is zero, producing no gradient update.

Summing Cases 1 and 3 yields $P(\text{correct sign or no update}) = p + (1-p)^K$. Since only Cases 1 and 2 produce non-zero gradients, the conditional probability follows as Equation 4.

Remark. Proposition 2 assumes binary episode rewards for analytical tractability. Under non-binary rewards (e.g., with per-step penalties for invalid actions), the group mean μ is nonzero even when all episodes fail, but the qualitative conclusion is preserved: advantage sign is determined by relative episode performance within the group, not by local action quality within individual windows. Section 4.4 validates this empirically using the full reward structure.

Penalty reward effect on credit agreement. Proposition 2 predicts 19.1% correct-sign rate (i.e., 80.9% incorrect) under pure binary reward. Our experiments include a -0.1 penalty per invalid action,

[2] The edge case where all K rollouts succeed (probability p^K) is negligible at the baseline success rates considered ($p^8 < 10^{-7}$ at $p = 12.5\%$).

making the reward non-binary. This penalty introduces variance within failed episodes: windows with valid actions receive $R_k = 0$ while windows with invalid actions receive $R_k = -0.1h$. Within a failed episode, the penalty-induced variance can occasionally produce positive $\hat{A}_k$ for windows with valid actions even when $\mu > 0$, since the normalization accounts for intra-group variance. This shifts the empirical credit agreement from the theoretical 19.1% toward 52%—still marginally above the 50% random baseline (p=0.025; +2pp effect size) and far below the level needed for convergence. A binomial test formally rejects $H_0: P = 0.191$ ($z = 45.9$, $p < 10^{-100}$), confirming that the gap between 19.1% (binary theory) and 52% (penalty-reward measurement) reflects the reward structure difference, not a failure of the theoretical framework.

We evaluate these expressions at the empirical baseline of our experiments. At $p = 12.5\%$ and $K = 8$: $(1-p)^K = 0.875^8 \approx 0.344$, giving $P(\text{correct or no update}) \approx 0.469$. Of the 65.6% of cases that produce a gradient update, only $P(\text{correct} \mid \text{update}) = 0.125/0.656 \approx 19.1\%$ have the correct sign. As K increases to 16, $(1-p)^K \approx 0.118$, and $P(\text{correct} \mid \text{update}) \approx 14.2\%$. In the large-$K$ limit, $P(\text{correct} \mid \text{update}) \to p$, so approximately $1 - p = 87.5\%$ of all gradient updates for locally correct windows point in the wrong direction.

A.3 Why the Community Defaults to Step-Independent Training

If step-independent training is structurally flawed for multi-turn agents, why has it become the default? We identify six contributing factors.

1. Context length constraints. Smaller language models (1B to 4B parameters) typically support context lengths of 2048 to 4096 tokens. Multi-turn episodes spanning 50 or more interaction steps easily exceed these limits. Step-independent windowing is a pragmatic response to a hardware constraint, not a deliberate algorithmic choice.

2. Token-level training inertia. Language model training is organized around predicting the next token over fixed-length sequences. When multi-turn RL inherits this paradigm, the natural unit of optimization becomes a fixed-length window rather than a variable-length episode. The mismatch between token-level optimization and episode-level credit assignment goes unquestioned because it aligns with existing training infrastructure.

3. Inherited framework default. The RAGEN framework (Wang et al., 2025) implements step-independent mode as its default configuration, not as one option among alternatives. Users inherit this choice without an explicit decision point, and default configurations propagate silently through downstream projects.

4. Memory and compute constraints. Episode-level training requires processing $O(T)$ tokens per sequence, incurring $O(T^2)$ attention cost with full self-attention. Step-independent training reduces this to $O(h)$ tokens and $O(h^2)$ attention cost per window. The computational savings are immediate and measurable, while the credit assignment failure is subtle and deferred until training collapse occurs.

5. Reward shaping as apparent fix. Dense reward shaping (per-step bonuses for intermediate goals) appears to address sparse reward problems. Our dense reward experiment (Section D) reveals a different failure mode: the agent maximizes format-level rewards (action validity rises from 60% to 97.5%) while task success remains at 0%. Reward shaping does not resolve sign misattribution; it creates new optimization targets that the agent can game without solving the underlying task.

6. Absence of negative results. When step-independent training fails, practitioners attribute the failure to insufficient hyperparameter tuning or the inherent difficulty of sparse rewards. Without published analyses that isolate the credit assignment mechanism as the root cause, the community cannot distinguish between "this problem is hard" and "this training paradigm is broken for this class of problems."

A.4 Connection to Classical Credit Assignment

Our findings connect to a long line of work on temporal credit assignment in reinforcement learning.

Temporal difference learning and eligibility traces. Sutton's TD(λ) (Sutton, 1988) provides a spectrum of credit assignment strategies parameterized by $\lambda \in [0, 1]$. At $\lambda = 0$, credit is assigned via one-step bootstrapped returns. At $\lambda = 1$, TD(λ) reduces to Monte Carlo returns. Step-independent training falls below this spectrum: windows are disconnected from each other, so temporal propagation of credit does not occur.

Generalized Advantage Estimation. GAE(λ, γ) (Schulman et al., 2016) extends TD(λ) with a separate discount factor. GAE(1, 1)

produces undiscounted Monte Carlo returns and is equivalent to episode-level GRPO without a critic. The key difference is that GAE uses value function subtraction as the baseline, while GRPO uses group normalization.

REINFORCE. Williams' REINFORCE algorithm (Williams, 1992) computes policy gradients from complete episode returns with a learned baseline $b(s)$. Episode-level GRPO is equivalent to REINFORCE where the group mean μ replaces $b(s)$ and the group standard deviation σ provides normalization. The algorithmic structure is not new. Our contribution is the diagnosis: the community's standard adaptation of this algorithm for multi-turn LLM agents (step-independent windowing) silently breaks the credit assignment structure that REINFORCE assumes.

Multi-agent credit assignment analogy. Step-independent windows can be understood through cooperative multi-agent games. Each window acts as an independent "agent" sharing a team reward but optimized without knowledge of other windows' actions. The sign misattribution problem parallels team reward credit assignment in cooperative multi-agent RL: when the team fails, all agents receive identical penalties regardless of individual contribution. In multi-agent RL, this is a recognized problem with known solutions such as counterfactual baselines and value decomposition methods.

B Hyperparameters

Table 3: Shared RL training hyperparameters across all conditions.

Parameter	Value
Group size K	8
Learning rate	1×10^{-6}
KL penalty β	0.01
Discount γ	0.95
Invalid action penalty	0.1
Tasks per training step	16
Training steps	15

All 1.7B experiments were conducted on a single NVIDIA A100 80 GB GPU. Each training run (15 steps, K=8 rollouts per task, 16 tasks per step) required approximately 6–8 GPU-hours. The 11-seed evaluation totaled ∼80 GPU-hours. The 8B experiment used 4× A100 80 GB with Fully Sharded Data Parallel (FSDP), requiring ∼24 GPU-hours per run. ScienceWorld experiments used the same single-GPU setup as the 1.7B ALFWorld runs.

SFT baseline. The base policy was fine-tuned on 2,546 expert demonstrations from the ALFWorld training split following the protocol of RAGEN (Wang et al., 2025), using learning rate 2×10^{-5}, batch size 16, and 3 epochs.

C Cross-Scale and Cross-Environment Results

Table 4: 8B results on ALFWorld. SFT and h=2 use decoding temperature τ=0; RL uses τ=0.4. Cell A: step 2 (post-collapse). Cell D: step 10. Single seed.

Method	In-Dist (%)	OOD (%)
SFT baseline	68.75	80.60
SFT h=2	31.25	—
Cell A: step-indep. (h=2)	28.1 → 1.5	—
Cell D: episode-level	**96.88**	**90.30**

D Additional Conditions

All conditions use single seeds unless noted.

Dense reward. Step-independent h=2 with +0.15 per valid action. Action validity rises from 60% to 97.5% while task success remains 0%; the agent games format-level rewards without solving tasks (Figure 6b).

REINFORCE++. Episode-level with per-token running-mean baseline (no critic). REINFORCE++ matches the SFT baseline (12.50%) without improvement, suggesting that per-token advantage normalization alone does not help when the per-token baseline lacks a learned value function. We note that 15 training steps may be insufficient for REINFORCE++ to converge; however, the Cell A collapse pattern represents active degradation rather than mere undertraining, as performance drops below the SFT initialization.

Discount factor sweep. $\gamma \in \{0.50, 0.80, 0.95, 1.00\}$ at h=5. All four collapse to 0% within 5 steps, including γ=1.0 where no magnitude decay exists. This confirms sign misattribution, not discounting, is the operative mechanism (Figure 6a).

Cell C attempt. Cell C (full observation + step-level advantage) would isolate M2 without M1. We attempted this configuration across six

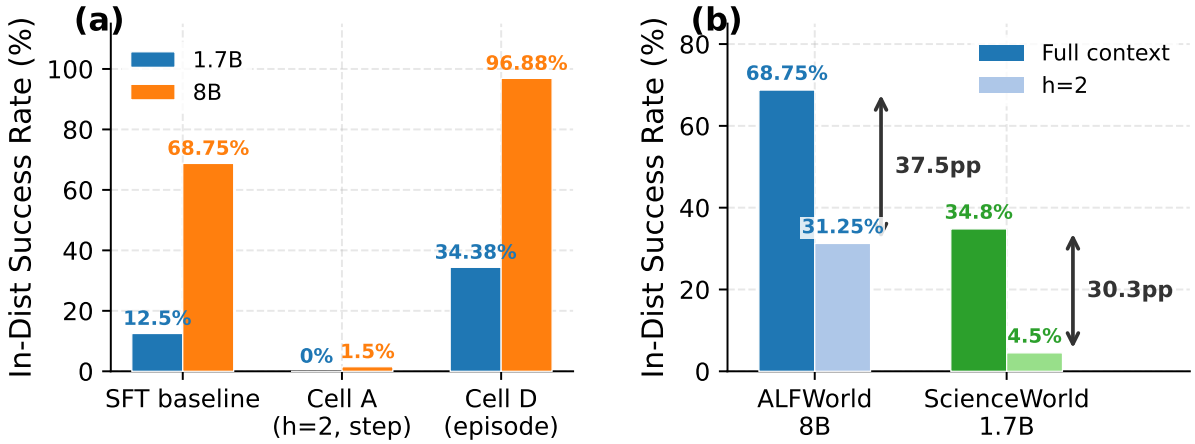


Figure 5: (a) Cross-scale verification (single seed): both collapse (Cell A) and fix (Cell D) are observed from 1.7B to 8B. The episode-level fix appears more effective at 8B (96.9% vs. 34.4% in-distribution). (b) Information loss (M1) quantification (single seed each): restricting context to h=2 drops SFT performance by 37.5pp at 8B ALFWorld and 30.3pp on ScienceWorld.

training runs with varying hyperparameters (KL penalty $\beta \in \{0.001, 0.01, 0.04\}$, learning rate $\in \{5\times10^{-7}, 10^{-6}\}$). All runs terminated due to training instability: KL divergence spiked above 10.0 within 2–3 steps, triggering NaN losses. We attribute this to the combination of full-length sequences (>4096 tokens) with step-level advantage normalization, which produces extreme advantage magnitudes at early window positions. Cell C remains an important gap in the factorial; addressing it likely requires architectural changes (e.g., gradient clipping per window position) beyond the scope of this diagnostic study.

E Response Length During Training

F Unified Step-5 Evaluation

To guard against cherry-picking of checkpoints, Table 5 reports all 11 seeds at a fixed evaluation point (training step 5). OOD results are included only for seeds with checkpoints evaluated under the D134 protocol (Section 4.1); remaining checkpoints were unavailable for post-hoc OOD evaluation.

Table 5: Episode-level GRPO: all 11 seeds at training step 5 (134 OOD tasks, 32 ID tasks, τ=0.4). SFT baseline: 12.5% ID, 6.72% OOD. OOD results marked "—" indicate checkpoints unavailable for post-hoc evaluation on the 134-task OOD set. ID results marked "—" indicate step-5 validation metrics not recorded in training logs.

Seed	Cat.	ID (%)	OOD (%)	vs SFT
seed2048	C	—	8.21	NS
seed8	C	28.13	11.94	NS
seed42	C	9.38	12.69	NS
seed200	S	—	10.45	NS
seed300	S	25.00	10.45	NS
seed100	S	31.25	11.19	NS
seed2	S	—	—	—
seed1	G[†]	—	23.88	p<0.001
seed4	G	—	—	—
seed5	G	—	—	—
seed6	G	21.88[‡]	—	—

[†] RAGEN built-in eval; checkpoint unavailable for re-evaluation.
[‡] Registry value 21.9% (7/32); rounded to 21.88% for consistency with 32-task denominator.

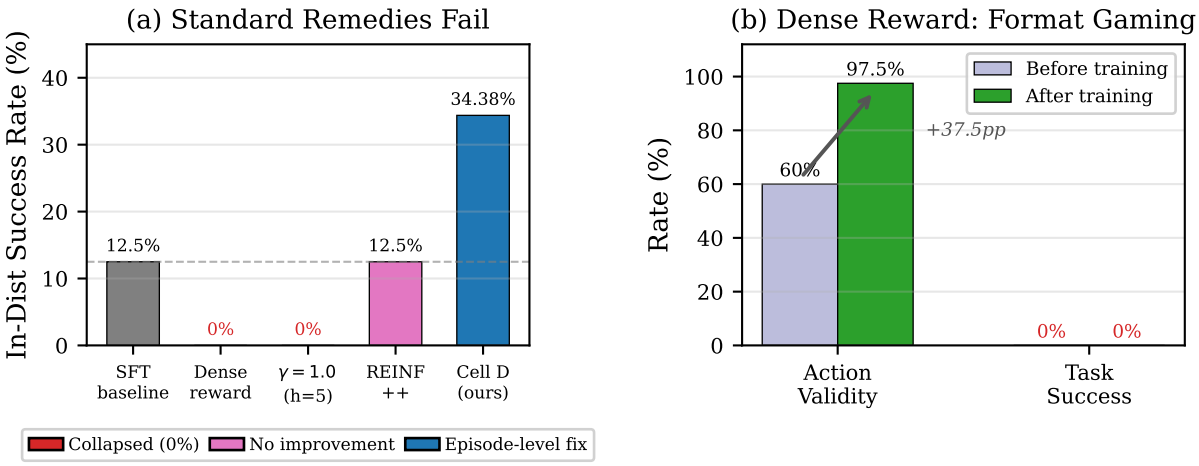


Figure 6: (a) Standard remedies for step-independent training: dense reward shaping and discount factor γ=1.0 both collapse to 0%; REINFORCE++ matches the SFT baseline without improvement. Only episode-level training (Cell D) exceeds baseline. (b) Dense reward gaming: action validity rises from 60% to 97.5% while task success remains 0%; the agent maximizes format-level rewards without solving tasks.

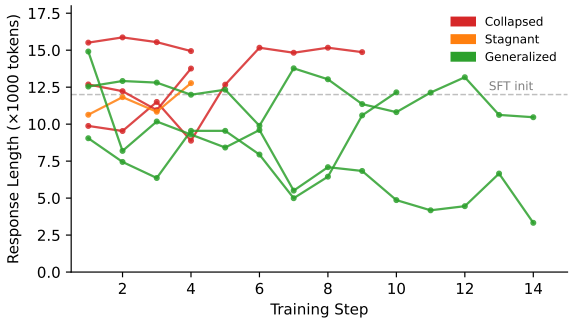


Figure 7: Response length over training for 7 seeds under episode-level GRPO. Collapsed seeds (red) exhibit length inflation to 14,000–16,000 tokens. Generalized seeds (green) maintain or reduce response length. Stagnant seed (orange, seed100) remains near the SFT initialization.